
SOLSHOT LITEPAPER

ARTILLERY COMBAT ON SOLANA

Version 1.0 // February 2026

[SolShot.gg](https://solshot.gg)

This document describes the design, mechanics, and tokenomics of SolShot, a browser-based multiplayer artillery game with Solana blockchain integration. It is not financial advice and does not constitute an offer of securities.

CONTENTS

- 01 OVERVIEW
- 02 GAME MECHANICS
- 03 WEAPON SYSTEM
- 04 GOLD ECONOMY
- 05 SOL WAGERING
- 06 SHOT TOKEN
- 07 PRESTIGE SYSTEM
- 08 SECURITY
- 09 ROADMAP
- 10 TEAM & LINKS

01 // OVERVIEW

SolShot is a browser-based, real-time multiplayer artillery game built on the Solana blockchain. Players compete in 1v1 tank battles, adjusting angle and power to fire projectiles across destructible terrain. The twist: matches can be wagered with real SOL, settled trustlessly through on-chain escrow smart contracts.

The game combines the nostalgia of classic artillery games (Pocket Tanks, Scorched Earth, Worms) with the speed and low cost of Solana transactions, creating a skill-based wagering experience where better players earn more.

KEY METRICS

0.01 - 0.5 SOL	90%	13	<2 sec
Wager Tiers	Winner Take	Launch Weapons	Settlement

CORE PRINCIPLES

Server-Authoritative. All physics, damage, and economy calculations run server-side. The client sends inputs (angle, power, weapon choice) and receives results. There is nothing to hack on the client.

Trustless Settlement. Wagers are held in on-chain Solana escrow programs. Neither the server nor any player can access escrowed funds. Settlement is automatic and verifiable.

Skill Over Spend. SHOT tokens are earned through gameplay milestones, not purchased. Prestige weapons offer variety, not power advantage. A new player with perfect aim beats a Diamond prestige player every time.

02 // GAME MECHANICS

MATCH FLOW

1. Both players connect Solana wallets and select a wager tier.
2. Both players deposit SOL into an on-chain escrow PDA.
3. The server generates a random terrain heightmap and assigns starting positions.
4. Players enter the weapon shop phase (30 seconds). Each starts with 1,000 Gold.
5. Combat begins. Players alternate turns, adjusting angle, power, and weapon selection.
6. The server calculates trajectories, impact, damage, and terrain deformation for each shot.
7. When one player reaches 0 HP, the round ends. Best-of-1, best-of-3, or best-of-5.
8. The winner receives 90% of the pot via on-chain settlement. 7% treasury, 3% operations.

MATCH MODES

MODE	PLAYERS	WAGER	FORMAT
Quick Match	1v1	0.01 - 0.1 SOL	B01 / B03
Duel	1v1	0.05 - 0.25 SOL	B03 / B05
High Roller	1v1	0.25 - 0.5 SOL	B03 / B05
Practice	1v1	Free	B01

TERRAIN

Terrain is represented as a 1D heightmap array generated server-side. Both clients receive identical terrain data. Explosions deform terrain by modifying the heightmap – creating craters, collapsing hills, and exposing buried tanks. Terrain is a core tactical element: high ground provides range advantage, valleys provide cover, and terrain manipulation weapons (Dirt Ball, Magic Wall) can reshape the battlefield.

WIND

A random wind value is assigned each round and displayed to both players. Wind affects projectile trajectory horizontally, requiring players to compensate. Wind direction and strength change between rounds, preventing memorized shot patterns.

03 // WEAPON SYSTEM

SolShot launches with 13 weapons across 6 tiers, plus 5 prestige-exclusive weapons unlocked by burning SHOT tokens. Each weapon has distinct physics behavior – there are no reskins or stat variations of the same weapon.

LAUNCH ROSTER (13 WEAPONS)

WEAPON	TIER	COST	BEHAVIOR
Single Shot	Free	0	Standard projectile, small blast. Infinite ammo.
Dirt Ball	Standard	150G	Raises terrain on impact. Defensive utility.
Magic Wall	Standard	200G	Erects terrain wall. Blocks incoming fire.
3 Shot	Tactical	400G	Three projectiles fan out mid-air.
Spider	Tactical	400G	Splits into crawling sub-munitions.
Heatseeker	Tactical	500G	Homes toward opponent tank.
Napalm	Rare	600G	Area burn, melts terrain.
Pile Driver	Rare	600G	Vertical drop, deep blast.
Sniper Rifle	Rare	700G	Tiny blast, precision damage.
Big Shot	Rare	700G	Large blast radius.
Jackhammer	Epic	1,000G	Drills into terrain, chain blasts.
Hail Storm	Epic	1,200G	Rains projectiles over wide area.
Crazy Ivan	Legendary	2,500G	15 random explosions. Total chaos.

PRESTIGE WEAPONS (5 EXCLUSIVE)

WEAPON	PRESTIGE	BURN COST	BEHAVIOR
Homing Missile	Bronze	200 SHOT	Guided to opponent. Reliable homing.
Chain Reaction	Silver	500 SHOT	Sequential blasts in chain.
Cruiser	Gold	1,200 SHOT	Player-guided missile. High skill ceiling.
Tommy Gun	Platinum	2,500 SHOT	Rapid-fire burst of small shots.
Mountain Mover	Diamond	4,000 SHOT	Massive terrain displacement. Ultimate.

Diamond prestige players have access to 18 weapons vs 13 for new players. This provides variety, not power advantage – a well-aimed Single Shot still beats a poorly-aimed Mountain Mover.

04 // GOLD ECONOMY

Gold is the in-match currency used to purchase weapons during the shop phase. It exists only within a match and never persists between matches. Gold cannot be traded, sold, or converted to any other currency.

EARNING GOLD

SOURCE	AMOUNT	NOTES
Starting Balance	1,000G	Every player, every match
Damage Dealt	+15G per HP	Incentivizes aggression
Kill Bonus	+200G	Finishing blow reward
Round Win	+300G	Winner of the round

BUDGET STRATEGY

With 1,000G starting balance, round 1 loadout options are constrained. Players can afford 2 Tactical weapons (400G each) plus 1 Standard (200G), or 1 Rare (700G) plus 1 Standard with Gold saved for round 2. The Legendary tier (Crazy Ivan at 2,500G) is impossible to buy in round 1 – players must earn Gold through combat first. This creates natural progression within each match.

05 // SOL WAGERING

WAGER TIERS

TIER	AMOUNT	TOTAL POT	WINNER (90%)	TREASURY (7%)	OPS (3%)
Micro	0.01 SOL	0.02 SOL	0.018 SOL	0.0014 SOL	0.0006 SOL
Low	0.05 SOL	0.10 SOL	0.09 SOL	0.007 SOL	0.003 SOL
Mid	0.10 SOL	0.20 SOL	0.18 SOL	0.014 SOL	0.006 SOL
High	0.25 SOL	0.50 SOL	0.45 SOL	0.035 SOL	0.015 SOL
Max	0.50 SOL	1.00 SOL	0.90 SOL	0.07 SOL	0.03 SOL

ESCROW MECHANISM

All wagers are handled by an Anchor smart contract deployed on Solana. The flow is:

1. Match creator calls `create_match`, which initializes an escrow PDA (Program Derived Address) with the wager amount and match parameters.
2. Both players call `deposit_wager`, transferring SOL from their wallets to the escrow PDA.
3. The match proceeds. The server tracks the authoritative match state.
4. On match completion, the server calls `settle_match` with the winner address. The escrow distributes funds: 90% to winner, 7% to treasury, 3% to operations.
5. If both players disconnect or the match is cancelled before completion, `cancel_match` returns deposits to both players.

All settlement uses integer lamport arithmetic (no floating point) to prevent rounding errors. Transactions are verified on-chain and settlement signatures are stored in the match record.

EDGE CASES

SCENARIO	OUTCOME
Player disconnects mid-match	30s reconnect window, then forfeit. Opponent wins pot.
Server crashes mid-match	Escrow refunds both players automatically.
Both players disconnect	Match cancelled. Escrow refunds both.
Timeout (no action 60s)	Turn auto-forfeits. After 3 forfeits, match ends.
Insufficient balance to join	Server rejects join. No escrow created.

06 // SHOT TOKEN

SHOT is the utility and governance token of the SolShot ecosystem. It is earned through gameplay milestones and burned permanently for prestige tier progression. SHOT is deflationary by design – the total supply can only decrease over time as players burn tokens for prestige.

TOKEN SPECIFICATIONS

PARAMETER	VALUE
Token Name	SHOT
Standard	SPL Token (Solana)
Total Supply	10,000,000 SHOT
Decimals	9
Mintable	No (fixed supply, mint authority burned)

DISTRIBUTION

ALLOCATION	AMOUNT	PERCENTAGE	VESTING
Reward Pool (PDA)	7,000,000	70%	Emitted via milestones, max 5%/mo
Treasury	1,500,000	15%	Controlled by multisig
Team	1,000,000	10%	12-month cliff, 24-month linear
Initial Liquidity	500,000	5%	Locked in Raydium pool

EMISSION MECHANICS

SHOT is emitted from the reward pool PDA based on gameplay milestones, not time. Examples of milestones: first match played, 10 wins, 100 matches, 1000 damage dealt in a single match, etc. The maximum emission rate is capped at 5% of the remaining pool per calendar month. As the pool depletes, emission rates naturally decrease – creating scarcity over time.

Critically, SHOT cannot be purchased directly. The only way to acquire SHOT is to play the game and hit milestones, or to buy it on the secondary market (Raydium). This ensures that SHOT distribution rewards active, skilled players.

BURN MECHANICS

When a player burns SHOT for prestige, the tokens are sent to a dead address and permanently removed from circulation. Burns are irreversible. The total SHOT supply decreases with every prestige action, making remaining SHOT more scarce over time. Total SHOT required to reach Diamond prestige: 8,400 SHOT (200 + 500 + 1,200 + 2,500 + 4,000).

07 // PRESTIGE SYSTEM

The prestige system rewards dedicated players with exclusive cosmetic upgrades and weapons. Each tier requires burning an increasing amount of SHOT tokens permanently. Prestige is a one-way commitment – once burned, tokens cannot be recovered.

TIER	BURN COST	CUMULATIVE	EXCLUSIVE WEAPON
Bronze	200 SHOT	200 SHOT	Homing Missile
Silver	500 SHOT	700 SHOT	Chain Reaction
Gold	1,200 SHOT	1,900 SHOT	Cruiser
Platinum	2,500 SHOT	4,400 SHOT	Tommy Gun
Diamond	4,000 SHOT	8,400 SHOT	Mountain Mover

Each prestige tier also unlocks cosmetic rewards: unique tank skins, kill effects, profile badges, and name borders. These are visual markers of commitment and have no gameplay advantage beyond the exclusive weapon unlocked at each tier.

Prestige is deliberately expensive. Reaching Diamond requires earning and burning 8,400 SHOT – representing hundreds of hours of gameplay. This ensures that high-prestige players are genuinely experienced, not just wealthy.

08 // SECURITY

SERVER AUTHORITY

SolShot uses a fully server-authoritative architecture. The client is a rendering layer only. All physics calculations (trajectory, impact, damage, terrain deformation), all economy operations (Gold earning, weapon purchases), and all match state transitions are computed server-side. The client sends only player inputs: angle, power, and weapon ID. The server validates these inputs, computes results, and broadcasts outcomes to both clients.

ANTI-CHEAT MEASURES

- Client inputs are validated against the current match state (correct turn, valid weapon, within bounds)
- Position data comes from the server, not the client – players cannot teleport or modify tank positions
- Gold balances and weapon inventories are tracked server-side – no client-side manipulation
- Rate limiting prevents spam actions and API abuse
- Settlement uses integer lamport arithmetic to prevent floating-point rounding exploits
- Async mutex locks on settlement operations prevent double-settlement race conditions

SMART CONTRACT SECURITY

The escrow program will undergo a third-party security audit before mainnet deployment. The program uses Anchor framework best practices: PDA-based escrow accounts, explicit account validation, and authority checks on all state-modifying instructions. The mint authority for SHOT will be burned after initial minting, making the token supply permanently fixed.

09 // ROADMAP

PHASE	TARGET	DELIVERABLES
1. Foundation	Weeks 1-2	MongoDB integration, weapon trimming, SolShot branding, server restructuring
2. Server Authority	Weeks 3-4	Server-side physics, turn validation, match state machine, Gold economy
3. Wallet + Wagering	Weeks 5-8	Wallet connect, escrow program, wager tiers, trustless settlement
4. SHOT + Prestige	Weeks 9-13	SHOT token mint, milestone emissions, prestige burns, exclusive weapons
5. Polish + Deploy	Weeks 14-16	UI reskin, art assets, mainnet deploy, Raydium pool, SolShot.gg live
6. Expand	Ongoing	Tournaments, leaderboards, terrain themes, seasonal content, mobile

10 // TEAM & LINKS

LINKS

Website: solshot.gg

Game: app.solshot.gg

Twitter: [@SolShotGG](https://twitter.com/SolShotGG)

Discord: discord.gg/solshot

GitHub: github.com/JJ-ME55/SolShot

DISCLAIMER

This litepaper is provided for informational purposes only. It does not constitute financial advice, an offer of securities, or a solicitation of investment. SHOT tokens are utility tokens with no expectation of profit. The value of SHOT may fluctuate and may go to zero. SolShot involves wagering real cryptocurrency – players should never wager more than they can afford to lose. Players must be 18+ to participate in wagered matches. SolShot may not be available in all jurisdictions. The team reserves the right to modify game mechanics, tokenomics, and roadmap as development progresses.

SOLSHOT // AIM. FIRE. EARN.